सत्यमेव जयते

Dr. Yogita Rana, I.A.S.
Joint Secretary
Ministry of Agriculture & Farmers Welfare
Government of India

D.O Letter No.14-33/2022-Fert Use

Dated 2nd February, 2024

Respected Sirs/Madams.

As you are aware that under Soil Health and Fertility Scheme, the Soil Sample test result are uploaded in Mobile Application manually to generate the Soil Health Card on the portal. It has been reported by some States during Video Conferencing that sometimes manual data enters to error. In order to avoid human interference and ensure the correct result entry, transforming conventional static soil testing laboratories into IoT-based facilities is the need of the hour.

2 The prime objective of this technology is to automate the labs, ensuring seamless and error-free operations, with the automated results feeding directly into the system. Since you are aware that this technology now days is inbuilt in the machine. However, most of the lab grounded is in past don't have this system. Therefore, it is requested to kindly undertake the transformation of the existing system. A SoP for transformation process in this regard is annexed.

3. I shall be grateful for your personal attention in the matter and directing the officer concerned to do the needful and take following steps to achieve the automation-

(i) Update the devices/ stages used in Soil Testing with IoT-based facilities to link with Mobile Apps to make automation, reduce the typographical errors and real-time monitoring in STLs.

(ii) The automation work may be done at your end with the help of M/s Krishi Tantra, Klonec Automation System Pvt. Ltd. Mangalore who is the web-portal maintainer of Soil Health Card. The point of contact in this matter will be Shri Sandeep Kondaji, CEO, Krishi Tantra, Mobile No. 06362380978.

Warm regards.

Yours Sincerely,

Yogita Rana

(Yogita Rana)

2/2/24

To,

The Additional Chief Secretaries/ Principal Secretaries, Agriculture, in all States/ UTs

Copy to:-

1. Directors (Agriculture) in all States / UTs
2. State Nodal Officers for Soil Health & Fertility Scheme

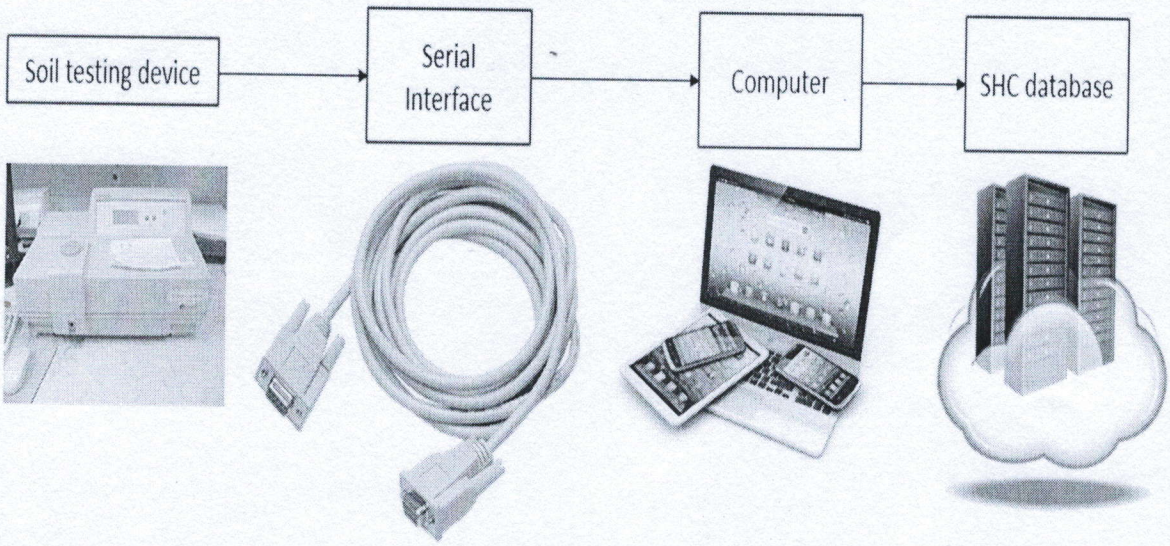
Standard Operating Procedure (SoP) for Transformation of Existing Device in Soil Labs

Transformation Process-

To achieve the above automation, following **transformation process** are brought to your kind notice-

i) Soil Testing Devices: All soil testing devices within the laboratory must be equipped with a serial communication port, eg. RS232, facilitating connectivity to a computer. Manufacturers are required to supply both device drivers and software necessary for retrieving results on a computer or mobile device.

ii) Serial interface Connection: To establish communication, the appropriate serial interface cables must be procured based on the laboratory's requirements. These cables will be used to connect the soil testing devices to the designated computer.



iii) Computer/Mobile integration: Upon completion of the soil testing process, the results will be retrieved using the software provided by the devices manufacturer. This software will act as an interface between the soil testing devices and the computer or mobile device, streamlining the data transfer process.

iv) SHC Database integration: The Soil Health Card team will collaborate with manufacturers by providing an API. This API will enable manufacturers to seamlessly integrate their software with the SHC database. After integration, manufacturers can upload the soil test results directly to the SHC database.

Benefits-

This comprehensive transformation process ensures a seamless flow of data from soil testing devices to the SHC database, leveraging standardized interfaces and connectivity solutions. The result is an automated and efficient system that enhances the accuracy and accessibility of soil health information.

The benefits of IoT-based STLs are as follows:

- IoT-based STLs will provide more accurate and up-to-date information, leading to better-informed agricultural decisions.
- Automation and real-time monitoring will increase the efficiency of soil testing procedures, saving time and resources.
- Automated systems allow for swift and real-time entry of results upon completion of testing.
- Automation eliminates the need for manual data entry, reducing the risk of typographical errors or inaccuracies associated with human input.